

A Model of the Factors Affecting Supply Chain Resilience: An Integrative Approach Incorporating Hesitant Fuzzy TOPSIS and Meta-Synthesis

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Abstract

A review of previous research shows that a systematic and comprehensive approach to the factors and dimensions of Supply

Chain Resilience (SCR) has say_farewell less attention in this field. Therefore, this study aims to design a model of factors influencing SCR. This research is considered applied and development-oriented in terms of its objective, and it was conducted in the tile and ceramic industry in Yazd province. In doing so, the study primarily investigated publications indexed in scientific databases, extracting the factors affecting SCR through a systematic approach and the meta-synthesis method. Following that, a model was constructed that consisted of 33 factors falling under five dimensions: 1) production and distribution management, 2) communication and participative management, 3) financial and information management, 4) human resources management, and 5) risk and crisis management. The findings revealed that such factors as "having a strong relationship with suppliers," "using lean production and eliminating wastes," "technological flexibility in production," "implementing integrated Information Technology (IT) infrastructures in the SC," and "safety stock of materials" were the most critical factors affecting resilience in the SC. Generally speaking, what distinguishes this study from others in this field is the new and scientific

framework of factors that it proposes through a systematic approach and the identification of the most important factors, which could serve as a guideline to senior managers and decision-makers.

Keywords: Supply chain, Resilience, Meta-synthesis, Hesitant fuzzy TOPSIS.

1|Introduction

In today's global economy, companies are trying to gain a competitive advantage and sustain their survival in the market, while customers are trying to find services/products that can meet their needs. Under such

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circumstances, Supply Chain (SC) management represents a significant and considerable element that can help respond to customers and gain a competitive advantage [1]-[3]. Chandra and Grabis [4] argue that an SC encompasses all stages that directly or indirectly fulfill customers' demands. From this viewpoint, a SC not only involves producers or suppliers but also incorporates transporters, wholesalers, retailers, and customers.

Moreover, as organizations operate in a volatile and unpredictable environment with inevitable disruptions in their SCs, their SCs must be designed to effectively and efficiently respond to any incidents. This includes the ability to restore the initial status or achieve an improved state after a disruption [5]. Disruptions are sudden and unexpected malfunctions that can stem from diverse causes, including natural disasters, fire, the loss of a crucial supplier, war, cyber-attacks, economic recession, sanctions, economic shocks, terrorism, and similar events [3], [5], [6]. If such incidents were to occur, they could lead to irreparable consequences for businesses [7]. These instances further underscore the significance of a resilient SC [8].

The resilient approach to SC management aims to empower an SC to revert to its original status with improved quality after analyzing a disturbance and avoiding failure scenarios [9]-[12]. Moreover, the nature of a resilient SC differs from that of a lean SC or an agile SC, bringing with it unique problems and challenges that must be accurately identified for the SC to maintain its competitiveness [13]. By relying on resilience, companies can effectively navigate SC failures and continue providing products/services to customers [14]. Recognizing the importance of this matter, employees at all corporate levels should possess some knowledge of failures and actively seek to learn from even minor setbacks.

In order to achieve specialization in management and effectively address both real and perceived risks, managers must establish a functional infrastructure to manage risks formally. This involves allocating human and informational resources [15]. Decision-making problems often arise in an environment of uncertainty, either due to limited human knowledge or decision-makers' restricted access to information. In such circumstances, experts should articulate their evaluations through linguistic statements (variables), creating a situation that necessitates the application of fuzzy logic [16]. However, experts often express hesitation in their choice of words when conveying opinions. Instead of providing a single membership degree, they use a set of values with corresponding membership degrees. To address this challenge, the Hesitant Fuzzy Set (HFS) approach has been introduced [17].

The purpose of this study is to propose a model of the factors affecting the Supply Chain Resilience (SCR) in the tile and ceramic industry in Yazd Province of Iran through the meta-synthesis approach and Hesitant Fuzzy Technique for Order of Preference by Similarity to Ideal Solution (HF-TOPSIS), to reach competitive advantage in unstable business conditions. One of the strategic industries that has left a remarkable impact on economic growth and employment in Yazd Province of Iran is the tile and ceramic industry; yet, a lack of SCR implemented in this industry could lead to irreparable losses when economic, political, and social crises break out in the province. On the other hand, the research conducted in the field of SCR indicates that each of them has approached the subject with a specific perspective. Some of the studies have identified and considered risk-related factors as necessary [18], [19]. Some other studies have focused on the financial management domain and paid attention to resilience from an economic standpoint [20]. Another category of studies has focused on the aspect of human resource management and the capabilities of organizational human resources in relation to resilience [21], [22]. Therefore, considering the variety of approaches to SCR and the lack of a comprehensive examination of the factors influencing its implementation, the present study employed a Meta-Synthesis approach to utilize the results of other studies and analyze them for the purpose of reaching a new summary.

Furthermore, due to the large number of factors, it is not feasible to compare them pairwise using methods such as AHP, BWM, etc. Therefore, the hesitant fuzzy TOPSIS method was employed for weighting and prioritizing the factors, which has been utilized in various studies [23]–[25]. Finally, since decision-making problems are inherently uncertain and we always face uncertainty in decision-making issues, the decisionmaker's selection of an option among linguistic variables is accompanied by a sense of doubt and hesitancy. Choosing only one value among these variables is very challenging and cannot fully reflect the decisionmaker's opinions. Therefore, the HFS theory, as a novel and powerful method for expressing decision-maker preferences, was utilized. It provides flexibility and allows for a comprehensive representation of the decisionmaker's preferences [26]. Considering the mentioned content, the present study aims to design a model for the factors influencing SCR in the tile and ceramic industry of Yazd province, in order to achieve a competitive advantage in an unstable business environment. The study employs a Meta-Synthesis approach and the hesitant fuzzy TOPSIS method to address the following questions:

- I. What are the factors influencing SCR in the tile and ceramic industry?
- II. How can the factors influencing the resilient SC in the tile and ceramic industry be prioritized?

2|Literature Review

SC management today plays a pivotal role in enhancing the performance of activities, particularly since the competitive dynamics among companies are fundamentally governed by their SCs [27]. In the past, the primary objective of SC design was to minimize costs or optimize services, although companies emphasize resilience today. The reason for this new tendency is that markets are now facing a higher degree of instability and chaos, and SCs are more vulnerable to disorders, especially as there are more risks to business survival now [28]. Meanwhile, SCs need more agility and flexibility to increase their responsiveness to demand fluctuations [29], [30].

Before defining the notion of resilience in the SC, it is essential to understand the nature of SC itself. A SC is an interconnected network of commercial partners engaged in production processes, transforming raw materials into final products or services to meet consumer demand [31], [32]. According to Carvalho et al. [33], an SC constitutes a specific framework of participants, their interrelationships, connections with other SCs, information and material flows, control management, and delivery timelines, all operating at a particular performance level within a specific time frame. The Council of Supply Chain Management Professionals (CSMP) defines SC management as the planning and coordination of all activities related to sourcing, logistics, exchange, and logistics management functions.

The initial definition of resilience was proposed by [34], who conceptualized it as a system's capacity to absorb changes and sustain its relationships. Pimm [35] perceives resilience as the speed at which a system restores equilibrium. Adger [36] was among the early proponents of the concept of "ecological resilience" and expanded its application to human societies by linking social resilience to social capital. In a broader context, resilience finds extensive application in various fields, including engineering, environmental sciences, and organizational research [37]. Numerous researchers endeavored to examine resilience in the context of SC, proposing conceptual frameworks for its understanding and implementation [38].

Resilience in the SC was initially introduced by [39]. It is the capability to respond to unexpected disturbances, including terrorist attacks or natural disasters, and to recover normal operations following such disruptions. Subsequently, Christopher and Peck [40] defined SCR as the ability to address risky events, revert to the previous status, and progress toward a new status. Following that, other researchers, such as Sheffi and Rice Jr [41], Berle et al. [42], and Pereira et al. [43], conceptualized SCR as the capability to recover from failure and overcome unexpected chaos. In essence, resilience refers to a firm's quick recovery from disorders in the SC.

Despite the development of definitions concerned with SC over time, most of the definitions suggest propositions that mostly build on the previous ideas or follow similar theoretical foundations [44]–[49]. Ponomarov and Holcomb [50] seem to have proposed the most theoretically comprehensive definition of resilience in the SC called the "adjustment capacity," which enables the SC to deal with unexpected incidents, respond to disruptions, and improve the situation while continuing the operations at an optimal level and controlling the structure and performance. In all of the ideas proposed about resilience, phrases are replaced with each other. However, the core concept denotes that resilience is the ability of a system to return to a stable status after a failure has occurred [51].

The research conducted in the field of SCR has primarily focused on four primary areas of development. These areas include 1) developing models or methods to examine the relationship between resilient SC and performance, as well as enhancing SC competitiveness, 2) assessing the level of compatibility of actions related to the resilient SC paradigm, 3) selecting the most suitable tools for a resilient SC, and 4) integrating SCR with other SC paradigms such as lean, agile, green, etc. As an example, Xiaoyu et al. [52] conducted a study using Interpretive Structural Modeling (ISM) and Fuzzy Set Qualitative Comparative Analysis (FSQCA) to examine the factors influencing SCR in 622 companies in China. They found that factors such as flexibility, agility, reconfiguration, visibility, and SC cooperation can have positive effects on SCR. In a study by Xie et al. [53], the researchers aimed to identify the critical factors affecting SCR in additive manufacturing architecture.

They conducted a literature review and expert interviews to identify seven dimensions and 21 indicators. Using the Intuitionistic Fuzzy Analytic Hierarchy Process (IFAHP), they concluded that flexibility is the most crucial factor in SCR. In a study by Samaranayake et al. [54], the researchers aimed to prioritize the driving factors of SCR using the fuzzy DEMATEL method. They found that among all the activities related to SC integration, information sharing with suppliers emerged as the most prominent factor for SCR. In a study by Liu et al. [55] on the factors influencing SCR in cross-border e-commerce, using fuzzy DEMATEL-ISM, they concluded that the following factors are vital influencers: speed of SC disruption recovery, response time to SC disruption, coordination capability among cross-border e-commerce partners, experience sharing among partners, joint planning among partners, establishing a good reputation for fairness within the organization, risk management team among partners, and fostering a culture of collaboration among partners. In their research titled "exploring SC network resilience in the presence of the ripple effect," Li and Zobel [56] focused on exploring strategies to create SCR by leveraging SC network structures. They emphasized the role of communication and transactions in enhancing the power of the SC. They proposed a multi-dimensional resilience measure to quantify resilient behaviors in the short, medium, and long term. Galetsi et al. [57] examined strategies for creating SCR from the perspective of Information Technology (IT) experts. They proposed allocating a larger budget towards infrastructure development as an effective strategy for enhancing SCR. This is because investing in infrastructure can lead to an increase in the value of SCs [57].

According to the existing research literature, necessary investigations and explorations have been conducted. Still, a study with the exact title that identifies and prioritizes the factors affecting resilient SC implementation using the hybrid meta-synthesis and HF-TOPSIS pproach was not found. This indicates a theoretical and empirical gap in this field. However, it is possible to utilize the available information in the relevant literature and conduct further investigations to identify and examine the factors influencing resilient SC implementation. If there is indeed a theoretical and empirical gap, this can be considered as an opportunity for further research in this area.

3|Research Methodology

This study constituted a descriptive survey aimed at fulfilling applied and developmental objectives. From a research duration perspective, the study was classified as cross-sectional research since it was conducted within a specific period that cannot be precisely replicated. The population in the first phase of the study comprised all publications indexed on authentic scientific databases. The selected publications focused on factors influencing the success of resilient SC as of the date when this study was conducted. In the second phase, ten individuals were selected from the population of all professors and managers specializing in SC within the tile and ceramic industry. This selection was conducted through purposive sampling, involving two professors and eight managers/experts in the field of tile and ceramic industry SCs. These experts provided their insights, contributing to the determination of the importance of each factor using the HF-TOPSIS method.

3.1|Hesitant Fuzzy Sets

The concept of HFS represents an extension of fuzzy sets where the membership function is defined across a range of possible values. This framework allows for the

effective representation of doubt and hesitation inherent in human judgments when dealing with decision-making problems. HFSs prove particularly valuable when faced with multiple ambiguous alternatives [58], [59]. Despite the significant utility of linguistic variables and their widespread application in decision-making scenarios, they also present certain limitations. In instances where decision-making problems entail uncertainty, selecting a single option from a set of linguistic variables involves a degree of doubt and hesitation. Under such circumstances, opting for just one value becomes exceedingly challenging and may fail to adequately capture the decision-makers perspectives [16], [60].

Based on the postulates of the HFS, hesitant fuzzy linguistic variables can serve as a new, flexible, and powerful way of expressing preferences in decision-making [60]. Through formulating a set of linguistic rules for extracting information, it would be possible to select several alternatives and express opinions through modifying phrases such as "lower than medium" or "between medium and good". Such statements help to overcome doubt in evaluations [16], [60].

3.2|TOPSIS for Hesitant Fuzzy Linguistic Term Sets

TOPSIS is one of the most frequently used methods for multi-criteria decision-making. In this approach, a positive ideal value and a negative ideal value are initially defined. Subsequently, the alternative exhibiting the least distance from the positive ideal and the greatest distance from the negative ideal is selected [61]–[65]. Xu and Zhang [66] introduced a multi-attribute TOPSIS for hesitant, fuzzy linguistic terms. The procedural steps in this method align with those of the traditional TOPSIS approach. Throughout all stages, including matrix aggregation, determination of the positive and negative ideals, and assessment of the distances between alternatives, the researchers relied on techniques specifically designed for hesitant, fuzzy linguistic terms. Essentially, Xu and Zhang's approach [66] builds upon the foundations of the classic TOPSIS method. Below, the steps of multiple-criteria decision analysis TOPSIS in a hesitant fuzzy environment are elucidated [66].

Step 1. Create the hesitant fuzzy decision matrix.

For an MCDM problem, the decision matrix $H = [h_{ij}]_{m \times n}$ is created, in which all h_{ij} ($i = 1, 2, 3 \dots m; j =$

$1, 2, 3 \dots, n$) values are hesitant fuzzy elements and the results of $A_i \in A$ is determined through $X_j \in X$ criteria, which are specified by decision-makers.

$$H = \begin{matrix} & \begin{matrix} A_1 & A_2 & \vdots & A_m \end{matrix} \\ \begin{matrix} h_{11} & h_{12} & \dots & h_{1n} \\ h_{21} & h_{22} & \dots & h_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ h_{m1} & h_{m2} & \dots & h_{mn} \end{matrix} \end{matrix} \quad (1)$$

Step 2. Calculate the weights of the criteria.

In this step, the weights of the criteria (which are the weights of the experts in this study) are determined using the maximizing deviation method, as described by Eq. (2).

$$= \frac{\sum_{j=1}^n \sum_{k=1}^m \sqrt{\frac{1}{N} \sum_{\lambda=1}^1 |h_{ij}^{\sigma(\lambda)} - h_{kj}^{\sigma(\lambda)}|^2}}{\sum_{j=1}^n \left(\sum_{i=1}^m \sum_{k=1}^m \sqrt{\frac{1}{N} \sum_{\lambda=1}^1 |h_{ij}^{\sigma(\lambda)} - h_{kj}^{\sigma(\lambda)}|^2} \right)} \quad (2)$$

Step 3. Determine the hesitant fuzzy positive ideal and negative ideal.

If A^+ is the hesitant, fuzzy, positive ideal and A^- is the hesitant, fuzzy, negative ideal, then we have

$$A^+ = \{h_{1+}, h_{2+}, \dots, h_{n+}\}. \quad (3)$$

$$A^- = \{h_{1-}, h_{2-}, \dots, h_{n-}\}, \quad (4)$$

where

$$h_j^+ = \bigcup_N \{h_{ij}\} = \left\{ \max_{\substack{k=1 \\ i=1}}^m \{h_{1j}^{\delta(k)}\} \right\}, j = 1, 2, \dots, n. \quad (5)$$

$$h_j^- = \bigcap_N \{h_{ij}\} = \left\{ \min_{\substack{k=1 \\ i=1}}^m \{h_{1j}^{\delta(k)}\} \right\}, j = 1, 2, \dots, n. \quad (6)$$

Step 4. Calculate the distance of each alternative from the hesitant fuzzy positive ideal and negative ideal.

In this step, the distance of all alternatives from the ideals is determined using the hesitant fuzzy Euclidean distance. The values of distances d_j^- and d_j^+ , which respectively indicate the distance from A^+ and A^- , are computed using Eq. (7) and Eq. (8).

$$d_{+i} = \sum_{j=1}^n W_j \left(\frac{1}{N} \sum_{k=1}^N (h_{ij}^{\delta(k)} - h_j^{+\delta(k)})^2 \right)^{\frac{1}{2}} \sum_{j=1}^n W_j d(h_{ij}, h_{+j}). \quad (7)$$

$$d_{-i} = \sum_{j=1}^n W_j d(h_{ij}, h_{-j}) = \sum_{j=1}^n W_j \left(\frac{1}{N} \sum_{k=1}^N (h_{ij}^{\delta(k)} - h_j^{-\delta(k)})^2 \right)^{\frac{1}{2}}. \quad (8)$$

Step 5. Calculate the relative closeness coefficient.

The relative closeness coefficient of the alternatives is computed via

$$C_i = \frac{d_{-i}}{d_{+i} + d_{-i}}, \quad (9)$$

where $0 \leq C_i \leq 1$, $i = 1, 2, \dots, m$; clearly, as C_i approaches 1, the alternative A_i becomes closer to the positive ideal solution (A^+) and becomes farther from the positive ideal solution (A^-).

Step 6. Rank the alternatives.

In the final stage, the alternatives $A_i (i = 1, 2, \dots, m)$ are ranked in ascending order based on their relative closeness coefficient values $C_i (i = 1, 2, \dots, m)$. The best alternative, which involves the greatest relative closeness coefficient value compared to the other alternatives, represents the best possible solution [66].

4|Research Findings

4.1|Extracting and Categorizing Success Factors in SCR through MetaSynthesis

This study used the seven stages proposed by Sandelowski et al. [67] to conduct the meta-synthesis method, as mentioned below:

Step 1. Organize the research question.

The question raised in Step 1 was "what are factors affecting the success of a resilient supply change."

Step 2. Conduct the systematic literature review.

In this step, a systematic search of resources published on databases was conducted using the relevant keywords, which included "resilience" or "resiliency" and "SC management" or "SC" or "SCM." As a result of this search procedure carried out on Scopus databases, a total of 698 publications were identified based on the keywords appearing in items such as titles, abstracts, and content. The search was conducted between 2000 and 2019.

Step 3. Search and find the relevant publications.

In this step, a certain number of publications were rejected during each revision, and they were subsequently excluded from the meta-synthesis procedure. The revision process in this study followed the stages depicted in Fig. 1.

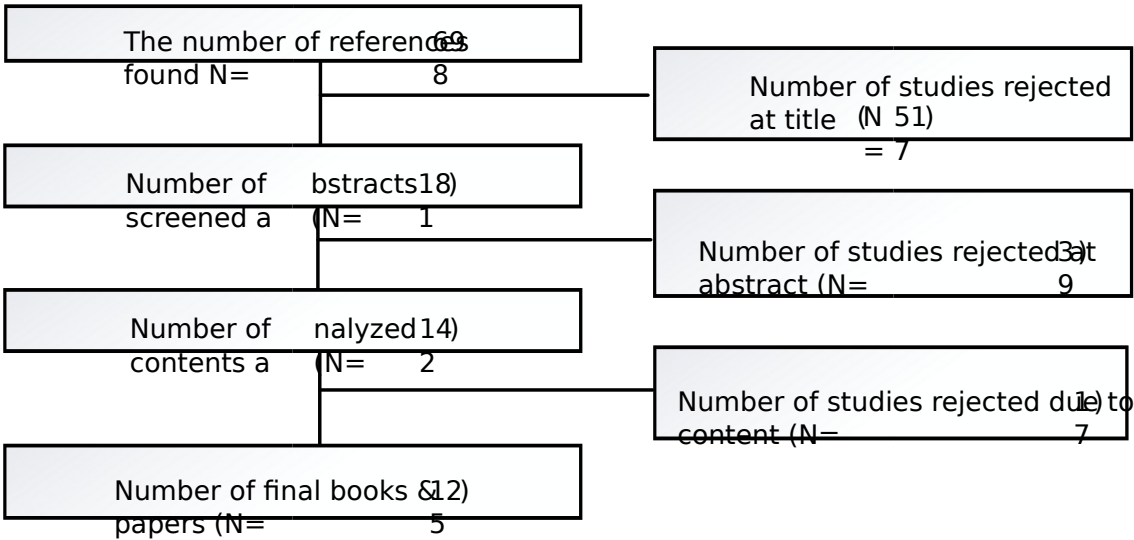


Fig. 1. Schematic summary of the search results and the selected resources.

As depicted in Fig. 1, a total of 125 publications were deemed relevant and acceptable during the search and selection process. Consequently, during the evaluation process, 573 papers were excluded from the initial 698, leaving a final selection of 125 publications for information analysis.

Step 4. Extract the results.

Throughout the entire meta-synthesis process, the contents of the ultimately selected publications were continuously revisited to extract clear insights into SCR

operations. The information obtained from these publications was categorized as follows: the reference details pertaining to each publication (including author name and year of publication) and the success factors of SCR management emphasized in each reference.

Step 5. Analyze and integrate qualitative findings.

The purpose of meta-synthesis is to offer a unified and novel interpretation of the findings. This methodology aids in clarifying concepts and identifying patterns within the accepted results. During the analysis, researchers search for topics and themes that can be found in the meta-synthesis method. Sandelowski et al. [67] call this stage "thematic investigation," through which themes and topics are recognized. When the topics and themes are detected, a category is shaped, and similar/relevant categories are put under themes that can perfectly describe them. The themes are foundations for creating explanations, models, theories, and working hypotheses. In this study, after extracting texts from the selected publications, overlaps were considered among them in terms of synonymity, resulting in the identification of 512 codes. Subsequently, 67 codes or concepts were distilled. Following this, the concepts contained within the codes were categorized under similar concepts, resulting in the identification of 33 factors. Once these factors were identified, they were grouped into five categories and considered as dimensions.

Step 6. Maintaining quality control.

In this study, the kappa coefficient was utilized to ensure quality control and assess the reliability of the model. Another expert in the field of SC management, who was unaware of how the codes and concepts were formulated in the research procedure, categorized the concepts independently. Subsequently, the categories identified in the research process were compared with those provided by the expert. As illustrated in *Table 2*, the researchers in this study proposed five groups, while the expert suggested six groups, with four categories being the same between the two. Based on the calculations, the kappa coefficient value was determined to be

0.56, indicating a good level of agreement. Consequently, the reliability of the model was confirmed.

Table 2. The ways the codes were converted into codes by the researchers and the expert.

		Researchers' Opinion		
		Yes	No	Total of First Codes
Expert's opinion	Yes	A=4	B=1	5
	No	C=2	D=0	2
	Total of second codes	6	1	N=7

$$\text{Chance agreement} = \frac{A+B}{N} \times \frac{A+C}{N} \times \frac{C+D}{N} \times \frac{B+D}{N} = \frac{5}{7} \times \frac{6}{7} \times \frac{2}{7} \times \frac{1}{7} = 0.0249.$$

$$\text{Observed disagreement} - \text{Chance agreements} = 0.571 - 0.0249 \text{ Kappa coefficient value} = 1 - \text{Chance agreements} = 1 - 0.0249 = 0.56.$$

Step 7. Reporting the findings.

In this step of the meta-synthesis method, the findings of the previous stages are presented in *Table 3*.

Table 3. Research model.

Row	Dimension	Factors
1	Production and distribution management	Technological flexibility in production
2		Flexibility in distribution
3		Using lean production and eliminating wastes
4		Employing the comprehensive quality management system
5		Offering productions compatible with customers' demands
6		Safety stock of materials
7		Safety stock of products
8		Having the ability to replace new markets
9		Offering new products continually and gradually
10	Communication and participative management	Interacting with social and environmentalist groups
11		Having a strong relationship with suppliers
12		Having a strong relationship with distributors
13		Being in contact with numerous suppliers
14		Making short-term and long-term contracts to respond to needs
15		Cooperating with competitors to predict joint security-related measures
16	Financial and informational management	Utilizing the knowledge management system
17		Identifying external opportunities/threats and formulating substitute plans
18		Dynamic pricing
19		Conducting financial/informational supervision to avoid disruptions
20		Sharing resources and information with competitors
21		Sharing resources and information of customers and distributors
22		Having access to credit and financial funds
23		Constructing integrated IT infrastructures in the SC

Table 3. Continued.

Row	Dimension	Factors
24	Human resources management	Using collaborative learning in the SC
25		Using experienced human resources for crisis

26		management
27		Employing multi-tasking teams Fostering a culture of risk-taking among employees and managers
28		Being able to recover the internal structure of the company rapidly
29		Being able to readily reconstruct the internal relations in the organization
30	Risk and crisis management	Sharing risks between SC partners
31		Identifying potential risk sources across the SC
32		Making quick responses to incidents and practicing crisis management
33		Creating new processes, strategies, and technologies to reduce risk

3.2|Ranking the Factors Affecting SCR threw Hesitant Fuzzy TOPSIS

After the factors affecting SCR were identified and categorized, copies of a questionnaire including 33 factors were submitted to 10 managers and experts in the field of SC in the tile and ceramic industry; more specifically, they were asked to determine the importance of each of the factors based on a 9-point scale ranging from "Very Unimportant" (0.1) to "Very Important" (0.9); in cases they were hesitant about an option, they could choose more than one choice. *Table 4* shows part of the results of this phase.

Table 4. The decision matrix for each expert.

	Expert 1	Expert 2		Expert 9	Expert 10
C1	(0.6, 0.7)	(0.6, 0.7, 0.8)		(0.6, 0.7, 0.8)	(0.6, 0.7, 0.8)
C2	(0.6, 0.7)	(0.6, 0.7, 0.8, 0.9)		(0.4, 0.5, 0.6)	(0.6, 0.7, 0.8, 0.9)
C3	(0.5, 0.6, 0.7, 0.8)	(0.6, 0.7, 0.8)		(0.6, 0.7, 0.8)	(0.6, 0.7, 0.8)
C4	(0.6, 0.7, 0.8, 0.9)	(0.4, 0.5, 0.6)		(0.3, 0.4)	(0.4, 0.5, 0.6)
C5	(0.6, 0.7, 0.8)	(0.6, 0.7, 0.8)		(0.5, 0.6, 0.7, 0.8)	(0.5, 0.6)
C6	(0.6, 0.7, 0.8, 0.9)	(0.2, 0.3, 0.4)		(0.5, 0.6)	(0.5, 0.6, 0.7, 0.8)
C7	(0.7, 0.8, 0.8)	(0.2, 0.3, 0.4)		(0.3, 0.4)	(0.6, 0.7, 0.8, 0.8)
C8	(0.5, 0.6, 0.7, 0.8)	(0.3, 0.4)		(0.4, 0.5, 0.6)	(0.4, 0.5, 0.6)
C9	(0.3, 0.4, 0.5, 0.6)	(0.5, 0.6)		(0.5, 0.6, 0.7, 0.8)	(0.6, 0.7, 0.8)
C10	(0.5, 0.6)	(0.4, 0.5, 0.6)		(0.2, 0.3, 0.4)	(0.4, 0.5, 0.6)
..	.	.	..	.	.
.	.	.	.	.	..
.	.	.	.	.	.
C33	(0.5, 0.6, 0.7, 0.8)	(0.6, 0.7, 0.8, 0.9)		(0.6, 0.7, 0.8)	(0.6, 0.7, 0.8)

Before proceeding to second phase, all hesitant fuzzy elements needed to be of the same length. In this study, the optimism principle was employed to determine the length of the elements. The maximum length among the hesitant fuzzy elements was found to be 4. To ensure uniformity, the greatest member was added to each hesitant fuzzy element, allowing them all to be of the same length. Consequently, the resulting decision matrix is presented in *Table 5*.

Table 5. The decision matrix with the same length of elements for each expert.

	Expert 1	Expert 2		Expert 9	Expert 10
C1	(0.6, 0.7, 0.7, 0.7)	(0.6, 0.7, 0.8, 0.8)		(0.6, 0.7, 0.8, 0.8)	(0.6, 0.7, 0.8, 0.8)
C2	(0.6, 0.7, 0.7, 0.7)	(0.6, 0.7, 0.8, 0.9)		(0.4, 0.5, 0.6, 0.6)	(0.6, 0.7, 0.8, 0.9)
C3	(0.5, 0.6, 0.7, 0.8)	(0.6, 0.7, 0.8, 0.8)		(0.6, 0.7, 0.8, 0.8)	(0.6, 0.7, 0.8, 0.8)
C4	(0.6, 0.7, 0.8, 0.9)	(0.4, 0.5, 0.6, 0.6)		(0.3, 0.4, 0.4, 0.4)	(0.4, 0.5, 0.6, 0.6)
C5	(0.6, 0.7, 0.8, 0.8)	(0.6, 0.7, 0.8, 0.8)		(0.5, 0.6, 0.7, 0.8)	(0.5, 0.6, 0.6, 0.6)
C6	(0.6, 0.7, 0.8, 0.9)	(0.2, 0.3, 0.4, 0.4)		(0.5, 0.6, 0.6, 0.6)	(0.5, 0.6, 0.7, 0.8)
C7	(0.7, 0.8, 0.8, 0.8)	(0.2, 0.3, 0.4, 0.4)		(0.3, 0.4, 0.4, 0.4)	(0.6, 0.7, 0.8, 0.8)
C8	(0.5, 0.6, 0.7, 0.8)	(0.3, 0.4, 0.4, 0.4)		(0.4, 0.5, 0.6, 0.6)	(0.4, 0.5, 0.6, 0.6)
C9	(0.3, 0.4, 0.5, 0.6)	(0.5, 0.6, 0.6, 0.6)		(0.5, 0.6, 0.7, 0.8)	(0.6, 0.7, 0.8, 0.8)
C10	(0.5, 0.6, 0.6, 0.6)	(0.4, 0.5, 0.6, 0.6)		(0.2, 0.3, 0.4, 0.4)	(0.4, 0.5, 0.6, 0.6)
C33	(0.5, 0.6, 0.7, 0.8)	(0.6, 0.7, 0.8, 0.9)		(0.6, 0.7, 0.8, 0.8)	(0.6, 0.7, 0.8, 0.8)

In this step, the weights of the experts were decided according to the maximizing deviations methods via *Eq. (2)* (see *Table 6*).

Table 6. The weights of each expert.

	Expert 1	Expert 2	Expert 3	Expert 4	Expert 5	Expert 6	Expert 7	Expert 8	Expert 9	Expert 10
W	0.0742	0.1045	0.1274	0.1007	0.0843	0.0925	0.1445	0.0783	0.1134	0.0803

Following that, based on *Eq. (5)* and *Eq. (6)*, the positive and negative ideals were calculated for each expert, as shown in *Table 7*.

Table 7. Positive and negative ideals for each expert.

	Positive Ideal				Negative Ideal			
Expert 1	0.7	0.8	0.8	0.9	0.3	0.4	0.5	0.6
Expert 2	0.6	0.7	0.8	0.9	0.2	0.3	0.4	0.4
Expert 3	0.8	0.9	0.9	0.9	0.1	0.2	0.2	0.2
Expert 4	0.8	0.9	0.9	0.9	0.2	0.3	0.4	0.4
Expert 5	0.6	0.7	0.8	0.9	0.2	0.3	0.4	0.4
Expert 6	0.8	0.9	0.9	0.9	0.4	0.5	0.6	0.6
Expert 7	0.8	0.9	0.9	0.9	0.1	0.2	0.2	0.2

Expert 8	0.8	0.9	0.9	0.9	0.2	0.3	0.4	0.4
Expert 9	0.8	0.9	0.9	0.9	0.2	0.3	0.4	0.4
Expert 10	0.8	0.9	0.9	0.9	0.4	0.5	0.6	0.6

In this step, the distance of each alternative from the positive ideal and the negative ideal was determined using *Eq. (5)* and *Eq. (8)*. Subsequently, the relative closeness coefficient was computed using *Eq. (9)*, and the results are presented in *Table 8*. As *Table 8* illustrates, criteria such as "having strong relationships with suppliers," "using lean production and eliminating wastes," "technological flexibility in production," "constructing integrated IT infrastructures in the SC," and "safety stock of materials" were the most critical factors affecting the implementation SCR.

Table 8. The distance of each indicator from the positive ideal and negative ideal and the relative closeness coefficient

Row	Indicators	d-	d+	Ci	Rank
1	Technological flexibility in production	0.4213	0.1035	0.8027	3
2	Flexibility in distribution	0.3015	0.2272	0.5703	8
3	Using lean production and eliminating wastes	0.4460	0.0798	0.8482	2
4	Employing the comprehensive quality management system	0.3683	0.1584	0.6993	6

Table 8. Continued.

Row	Indicators	d-	d+	Ci	Rank
5	Offering productions compatible with customers' demands	0.2518	0.2774	0.4757	12
6	Safety stock of materials	0.3686	0.1593	0.6983	5
7	Safety stock of products	0.1775	0.3488	0.3372	26
8	Having the ability to replace new markets	0.2232	0.3016	0.4253	17
9	Offering new products continually and gradually	0.2421	0.2858	0.4586	10
10	Interacting with social and environmental groups	0.0984	0.4280	0.1870	32
11	Having a strong relationship with suppliers	0.4524	0.0710	0.8643	1
12	Having strong relationships with distributors	0.3264	0.1994	0.6208	7
13	Being in contact with numerous suppliers	0.1896	0.3381	0.3593	24
14	Making short-term and long-term contracts to respond to needs	0.2217	0.3091	0.4177	14

15	Cooperating with competitors to predict joint security-related measures	0.245 9	0.286 8	0.461 6	11
16	Employing the knowledge management system	0.158 7	0.365 4	0.302 8	30
17	Identifying external opportunities/threats and formulating substitute plans	0.188 6	0.336 0	0.359 4	19
18	Dynamic pricing	0.2229	0.302 8	0.424 0	13
19	Conducting financial/informational supervision to avoid disruptions	0.262 7	0.263 7	0.499 1	16
20	Sharing resources and information with competitors	0.125 1	0.401 1	0.237 7	29
21	Sharing resources and information of customers and distributors	0.183 2	0.342 7	0.348 3	23
22	Having access to credit and financial funds	0.154 2	0.371 6	0.293 3	28
23	Constructing integrated IT infrastructures in the SC	0.415 8	0.111 3	0.788 8	4
24	Using collaborative learning in the SC	0.126 9	0.401 1	0.240 4	31
25	Using experienced human resources for crisis management	0.260 7	0.265 4	0.495 5	9
26	Employing multi-tasking teams	0.230 7	0.295 3	0.438 6	18
27	Fostering a culture of risk-taking among employees and managers	0.180 8	0.347 7	0.342 0	22
28	Being able to recover the internal structure of the company rapidly	0.198 1	0.326 1	0.378 0	21
29	Being able to readily reconstruct the internal relations in the organization	0.187 7	0.338 2	0.356 9	20
30	Sharing risks between SC partners	0.176 7	0.351 3	0.334 6	27
31	Identifying potential risk sources across the SC	0.189 5	0.335 2	0.361 1	25
32	Making quick responses to incidents and practicing crisis management	0.218 8	0.308 6	0.414 9	15
33	Creating new processes, strategies, and technologies to reduce risk	0.304 4	0.220 1	0.580 3	8

4|Conclusion

This study drew on the meta-synthesis and HF-TOPSIS methods and the opinions of industrial and academic experts to design a model of the factors affecting the success of a resilient SC. Based on the research literature and background investigation, using the Meta-Synthesis approach identified 33 success factors for resilient SC in the tile and ceramic industry across five dimensions: production and distribution management, communication and collaboration management, financial and information management, human resource management, and crisis and risk management. Subsequently, the study employed the HF-TOPSIS method to prioritize the factors influencing SCR in the tile and ceramic industry. According to the results, strong supplier relationships, the use of lean production and waste elimination, and technological flexibility in production were identified as the most critical success factors for resilient SC implementation. Considering the theoretical works addressing the topic and its literature, one learns that the findings of this

study are supported by the observations of other studies [7], [33], [44], [47], [49], [68]–[72].

Many other studies have emphasized "having a strong relationship with suppliers" as one of the most significant factors in successfully implementing a resilient SC. This factor plays a crucial role in creating synergy, reducing costs, and creating a flexible supply capacity. Furthermore, "having a strong relationship with suppliers" could lead to reduced lead time, lower costs, and higher product quality. Such elements would enhance the adjustment capacity of the SC in the face of unexpected incidents and response to disruptions. The second most crucial factor in constituting a resilient SC was "using lean production and eliminating wastes"; Pettit et al. [49], Mensah and Merkuryev [32], Wang et al. [73], and Blackhurst et al. [45] also found lean production as one of the most influential factors.

In general, lean production serves as a strategy aimed at eliminating waste and fostering continuous improvement across all production processes. Researchers assert that implementing lean production principles within the SC can enhance product efficiency, reduce costs, boost profitability, and increase flexibility. This heightened flexibility and efficiency, in turn, bolster the SC's capacity to recover from failures and navigate unforeseen disruptions. "Technological flexibility in production" was another vital factor and critical element contributing to the formation of a resilient SC. This indicator would remarkably contribute to SCR's adjustment to required changes within the shortest period and with minimum effort. For this reason, many researchers have highlighted this indicator in their investigation [32], [33], [49], [68], [71], [74]–[81].

"Constructing integrated IT infrastructures in the SC" emerges as another highly significant success factor in resilient SCs, as highlighted in studies by [71], [72], [82], [83]. By creating such informational and communicational infrastructures, information could be shared in all of the areas of the SC, thus enhancing the clarity of performance across the SC for the partners involved. Therefore, senior managers must formulate necessary plans to construct IT infrastructures to handle SC disruptions. Another factor that clearly affects SCR is "safety stock" (redundancy), as emphasized by many researchers such as Christopher and Peck [40], Briano et al. [84], Zsidisin and Wagner [75], Karl et al. [85], Azadeh et al. [76], Bevilacqua et al. [72], Tang [86], Fakoor et al. [80], Sáenz and Gutiérrez [87], Blackhurst et al. [45], and Carvalho et al. [33].

Any organization must consider "safety stock" as a strategy to deal with disruptions in the production process. Managers of tile and ceramic companies are encouraged to implement a range of measures outlined above to enhance the resilience of their SCs and improve their capacity to respond promptly when disruptions occur. Future studies could extend this research to other industries to yield more comprehensive and generalizable results, leading to the development of a more comprehensive model. Researchers may also utilize the model and its factors to compare and rank tile and ceramic companies based on their resilience. Employing quantitative survey methods such as cluster analysis to code and categorize dimensions could prove beneficial. Additionally, categorizing the factors could be accomplished through exploratory and confirmatory factor analysis.

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Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

No potential conflict of interest was reported by the author(s).

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